House Price Prediction – Project Summary

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Project Objective

The objective of this project was to develop a machine learning model for predicting house prices using various property characteristics. The project involved data exploration, preprocessing, model training, evaluation, and visualization to understand the factors influencing house prices.

Data Preparation and Preprocessing

The dataset was analyzed using Pandas to identify missing values and duplicate records. Categorical variables were converted into numerical form using one-hot encoding, and all relevant features were retained for model development. The dataset was then divided into training and testing sets for evaluation.

Models Implemented

Two regression models were developed using Scikit-learn:

- Linear Regression
- Random Forest Regressor

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Key Findings

Linear Regression achieved better performance with an R^2 score of approximately 0.65, indicating that the model was able to explain around 65% of the variation in house prices. Features such as area, number of bathrooms, parking spaces, number of stories, and air conditioning were found to have the greatest influence on property prices. A notable observation was that the Random Forest model did not outperform Linear Regression on this dataset.

Conclusion

This project demonstrated the practical application of machine learning in real estate price prediction. The analysis suggests that property size and amenities significantly impact house prices. These insights can help real estate businesses make better pricing decisions and understand the factors that contribute most to property value.